

Live EDA Assistant - Automated Report

Dataset: tested.csv

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1. Dataset Overview

Rows: 418

Columns: 12

Missing Cells: 414 (8.25%)

Duplicate Rows: 0

Overview Insights:

- Dataset has low missing values (8.3%), which is good for analysis.

2. Column-Level Insights

- In categorical column 'Name', value 'Kelly, Mr. James' is most frequent (0.2% of rows).
- In categorical column 'Sex', value 'male' is most frequent (63.6% of rows).
- Column 'Age' has noticeable missing values (20.6%). Consider imputing values using mean/median/mode.
- Numeric column 'SibSp' is highly right-skewed (skewness=4.17). Applying transformations (log/box-cox) may help.
- Numeric column 'Parch' is highly right-skewed (skewness=4.65). Applying transformations (log/box-cox) may help.
- Column 'Parch' has many outliers (22.5% of values). Outlier handling (capping/removal) might improve model robustness.
- In categorical column 'Ticket', value 'PC 17608' is most frequent (1.2% of rows).
- Numeric column 'Fare' is highly right-skewed (skewness=3.69). Applying transformations (log/box-cox) may help.
- Column 'Fare' has many outliers (13.2% of values). Outlier handling (capping/removal) might improve model robustness.
- Column 'Cabin' has very high missing values (78.2%). It might be a candidate to drop or replace with a proxy feature.
- In categorical column 'Cabin', value 'B57 B59 B63 B66' is most frequent (3.3% of rows).
- In categorical column 'Embarked', value 'S' is most frequent (64.6% of rows).

3. Correlation Insights

- Columns 'Pclass' and 'Fare' are strongly negatively correlated (corr=-0.58). They may carry similar information.
- Columns 'Pclass' and 'Age' are strongly negatively correlated (corr=-0.49). They may carry similar information.
- Columns 'Age' and 'Fare' are strongly positively correlated (corr=0.34). They may carry similar information.
- Columns 'SibSp' and 'Parch' are strongly positively correlated (corr=0.31). They may carry similar information.

4. Outlier Summary (Preview)

- PassengerId: 0 outliers (0.0% of values), normal range $\approx$ [683.50, 1517.50]
- Survived: 0 outliers (0.0% of values), normal range $\approx$ [-1.50, 2.50]
- Pclass: 0 outliers (0.0% of values), normal range $\approx$ [-2.00, 6.00]
- Age: 2 outliers (0.6% of values), normal range $\approx$ [-6.00, 66.00]
- SibSp: 11 outliers (2.6% of values), normal range $\approx$ [-1.50, 2.50]
- Parch: 94 outliers (22.5% of values), normal range $\approx$ [0.00, 0.00]
- Fare: 55 outliers (13.2% of values), normal range $\approx$ [-27.51, 66.91]